🔑 **Answer Key — Teacher Copy · Lesson 7-5**

MSTAR PRACTICE WORKSHEET

Lesson 7-5: Represent Rational Numbers on the Coordinate Plane

Grade 6 Mathematics · Standard 6.NOS.6

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

Which ordered pair names a point 3 units right and 7 units up from the origin?

A (0, 7)

B (3, 7)

C (3, 3)

D (7, 3)

The first coordinate is horizontal (right 3), the second is vertical (up 7). So (3, 7).

Question 2 SELECTED RESPONSE — CORRECT: A

What are the coordinates of the origin?

A (0, 0)

B (0, 1)

C (1, 1)

D (1, 0)

The origin is where the x-axis and y-axis cross, at (0, 0).

Question 3 SELECTED RESPONSE — CORRECT: A

A point is at (6, 2). What does the 6 tell you?

- (A) Move 6 units to the right
- (B) Move 6 units up
- (C) Move 6 units to the left
- (D) The point is in Quadrant 6

> The first number (x-coordinate) tells you horizontal movement. 6 means go right 6 units from the origin.

Question 4 SELECTED RESPONSE — CORRECT: C

Spot 1 — Captain Vega's first clue says the Sandy Dock is right 7 and up 2 from the origin. Which ordered pair names the Sandy Dock?

- (A) (2, 7)
- (B) (2, 2)
- (C) (7, 2)
- (D) (7, 0)

> The x-coordinate is the sideways move (right 7) and the y-coordinate is the up move (up 2), so the Sandy Dock is (7, 2).

Question 5 SELECTED RESPONSE — CORRECT: A

Spot 2 — The map marks the Lookout Tower at (4, 9). What does the 4 tell you to do from the origin?

- (A) Move 4 units to the right along the x-axis
- (B) Move 4 units up along the y-axis
- (C) Move 4 units to the left
- (D) Stay on the y-axis at 4

> The first number in (x, y) is the x-coordinate, so 4 means move 4 units to the right along the x-axis before moving up.

Question 6 SELECTED RESPONSE — CORRECT: B

Spot 3 — Vega writes: 'The Coconut Grove is 3 across and 8 up.' Which point on the map is the Coconut Grove?

A (0, 8)

B (3, 8)

C (3, 3)

D (8, 3)

Across (3) is the x-coordinate and up (8) is the y-coordinate, so the Coconut Grove is at (3, 8).

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

Starting at the origin, which point is located by moving 4 units LEFT and 3 units UP?

A (3, −4)

B (−3, 4)

C (−4, 3)

D (4, 3)

An ordered pair is (x, y). Moving left makes x negative, so x = −4; moving up makes y positive, so y = 3. The point is (−4, 3).

- **A:** The coordinates are swapped and the signs with them. The x-coordinate comes first.
- **B:** The two numbers are swapped. Four left gives x = negative 4 and three up gives y = 3.
- **D:** Moving LEFT makes x negative. Four units left of the origin gives x = negative 4.

Part B — correct answer: B

Why do (−4, 3) and (3, −4) name different points?

A They are actually the same point drawn twice.

B The order matters: the first number is always the horizontal move, the second the vertical.

C Negative coordinates cannot come first.

D The larger number is always written first.

An ordered pair is ordered, and that is not a play on words. (−4, 3) is 4 left and 3 up; (3, −4) is 3 right and 4 down — different quadrants entirely.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about plotting points:

A The origin is the point (0, 0).

B The x-coordinate is written first in an ordered pair.

C A point on the y-axis has an x-coordinate of 0.

D The order of the two coordinates does not matter.

E Moving down from the origin makes the y-coordinate negative.

A, B, C, and E are true. D is false — swapping the coordinates usually names a completely different point.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A student says that (3, 5) and (5, 3) are the same point because they use the same numbers. Is the student correct? Explain why or why not, and describe where each point is located on the coordinate plane.

Model answer: The student is not correct. In (3, 5) the 3 is the x-coordinate, so you move right 3 and up 5, landing high and toward the left. In (5, 3) the 5 is the x-coordinate, so you move right 5 and up 3, landing lower and farther right. Same digits, different jobs — the order inside an ordered pair decides which direction each number controls.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.
